

Degradable polyurethane for marine anti-biofouling

Chunfeng Ma,^a Liguu Xu,^a Wentao Xu^a and Guangzhao Zhang^{*ab}Cite this: *J. Mater. Chem. B*, 2013, **1**, 3099

Degradable polyurethane (PU) with copolyester oligomer consisting of ϵ -caprolactone (CL) and glycolide (GA) as the soft segments has been prepared by a combination of ring-opening polymerization and condensation reaction. Enzymatic and hydrolytic degradation experiments demonstrate that the PU can degrade in seawater. Such a polyurethane exhibit a more rapid degradation in comparison with that with poly(ϵ -caprolactone) (PCL) soft segments because the introduction of GA can reduce the crystallinity, as revealed by differential scanning calorimetry (DSC) and polarizing optical microscope (POM). Marine field tests show that the degradable polyurethane has good antifouling ability due to its self-renewal property. Besides, such polyurethane can serve as a carrier and controlled release system for an antifoulant, and the incorporation of an antifoulant in the polyurethane can significantly improve the antifouling ability and duration.

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Introduction

Marine biofouling worldwide happens due to the undesirable accumulation of microorganisms, plants and marine animals on submerged surface. It is a serious problem for maritime industries and other marine activities.^{1–3} Marine submerged surfaces are usually coated with antifouling materials to prevent or control biofouling. Biocidal coatings containing tributyltin had been the most effective ones until they were banned because they are detrimental to non-target organisms and the surrounding environment. Currently, the most used antifouling materials contain cuprous oxide (Cu_2O) together with some organic biocides, which are also ecologically harmful.^{4,5} Accordingly, some efforts have been turned to the development of environment-friendly materials such as low surface energy elastomers,^{6–8} bioinspired engineered topographies,^{9,10} amphiphilic nanostructured coating,^{11–15} enzymes,^{16,17} and some antifoulants made of natural products.^{18,19} However, their long term properties in marine environments needs testing.

In principle, a surface constructed from a biodegradable polymer would be gradually decomposed and eroded under seawater or enzymatic attack in marine environments, which can polish the attached living organisms or inorganics. In other words, the degradation can lead to a self-renewal surface. Actually, we have showed a degradable polymer with protein resistance that can effectively inhibit marine biofouling.²⁰ On the other hand, considering that biodegradable polymers can be used for the delivery and controlled release of drugs,^{21,22} they

are able to facilitate antifoulant release. Some attempts have been made to use poly(ϵ -caprolactone) (PCL) and its copolymers as the carrier of antifoulant (Cu_2O),^{23,24} and the bio-surface erosion of the PCL/polyhydroxyalkanoates (PHAs) blends by microbial degradation was also examined.²⁵ However, the low degradation rate and low adhesion strength limit their applications.

In the present study, we have prepared dihydroxyl-terminated copolyester oligomers CL/GAx consisting of CL and GA, where the introduction of GA is to moderate the crystallinity and improve the biodegradation. The polyaddition of oligomer with L-lysine ethyl ester diisocyanate (LDI) and 1,4-butanediol (BDO) yields a biodegradable polyurethane (CL/GAx-PU) with good adhesion. The enzymatic degradation of the PU has been investigated by quartz crystal microbalance with dissipation (QCM-D), and the anti-biofouling ability of the degradable PU with or without antifoulant is evaluated by marine field tests. Our aim is to develop non-toxic, effective and durable anti-fouling material.

Experimental section

Materials

ϵ -Caprolactone (CL) (Aladdin) was dried over CaH_2 for 48 h and distilled under reduced pressure prior to use. Glycolide (GA) (Daigang Biomaterial) was recrystallized from ethyl acetate three times. Diethylene glycol (DEG) and 1,4-butanediol (BDO) from Guangzhou Chemical Reagent Factory were purified by distillation under reduced pressure. L-lysine ethyl ester diisocyanate (LDI) from Dahong Chemical was used as received. Dibutyltin dilaurate (DBTDL) and stannous octoate ($\text{Sn}(\text{Oct})_2$) were from Aladdin and used as received. Tetrahydrofuran (THF) was refluxed over CaH_2 and distilled prior to use. Lipase PS purchased from Aldrich was purified by

^aFaculty of Materials Science and Engineering, South China University of Technology, Guangzhou, 510640, P. R. China. E-mail: gzzhang@ustc.edu.cn

^bHefei National Laboratory for Physical Sciences at Microscale, Department of Chemical Physics, University of Science and Technology of China, Hefei, 230026, P. R. China

filtration and freeze-drying before use. 4,5-Dichloro-2-octylisothiazolone (DCOIT) was kindly presented by Fangfu Chemical Co. (Guangzhou). Artificial seawater was prepared following ASTM D1141-98 (2003). Other reagents were used as received. The synthesis of CL/GAx and CL/GAx-PU is illustrated in Scheme 1.

Synthesis of CL/GAx macrodiol

The dihydroxyl-terminated oligomer with different CL/GA ratios were synthesized by ring-opening polymerization of CL and GA monomer by using DEG and Sn(Oct)₂ as the initiator and catalyst, respectively. Typically, ϵ -CL (5.25 g, 46 mmol), GA (0.109 g, 0.9 mmol), DEG (0.3 g, 2.8 mmol) and Sn(Oct)₂ (0.01 g, 0.2 wt% based on monomers) were introduced into a 25 mL three-neck flask under a nitrogen purge and the polymerization was carried out at 125 °C for 24 h. The products were dissolved in THF and precipitated in *n*-hexane and then dried in a vacuum oven for 24 h. ¹H NMR (400 MHz, CDCl₃, ppm): 4.7 (COCH₂O), 4.1 (COCH₂CH₂CH₂CH₂CH₂O), 3.4 (OCH₂), 2.3 (COCH₂CH₂CH₂CH₂CH₂O), 1.6 (COCH₂CH₂CH₂CH₂CH₂O), 1.4 (COCH₂CH₂CH₂CH₂CH₂O). FTIR: 3500 cm⁻¹ (OH), 2860–2950 cm⁻¹ (CH₂), 1725 cm⁻¹ (C=O). The macrodiol is designed as CL/GAx, where *x* is the molar percentage of GA determined by ¹H NMR (Table 1).

Synthesis of CL/GAx-PU

The polyurethane was synthesized *via* a two-step condensation reaction in THF under a nitrogen atmosphere.^{26,27} The reaction of LDI with CL/GAx macrodiol at 80 °C for 2 h yielded a pre-polymer. Subsequently, 1,4-BD and DBT were added as the chain extender and catalyst, respectively, and the mixture was stirred at 90 °C for 12 h. The resultant polymer was precipitated in *n*-hexane and vacuum-dried for 24 h. The soft segment consists of PCL, CL/GA2, CL/GA6 or CL/GA10, and the soft segment accounts for 80 wt% of the PU. The characterization data are shown in Table 1.

Characterizations

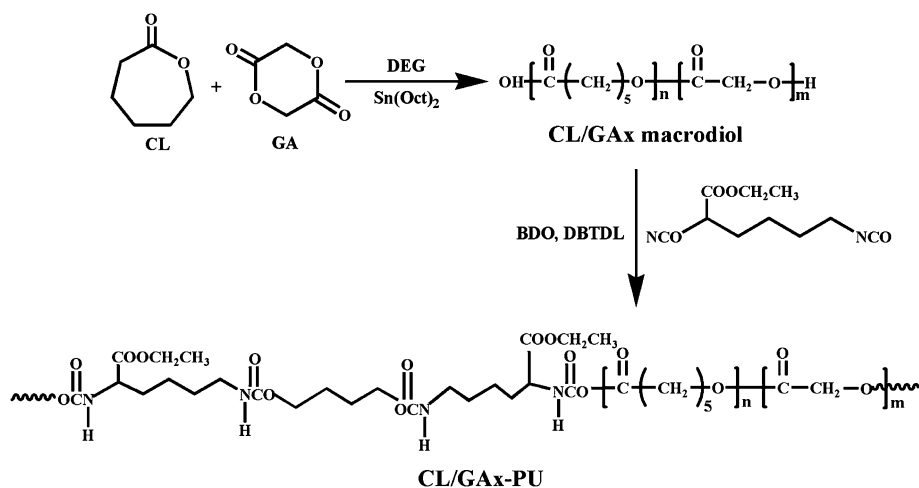
Proton nuclear magnetic resonance spectroscopy (¹H NMR). All ¹H NMR spectra were recorded on a Bruker AV400 NMR instrument with CDCl₃ as solvent and tetramethylsilane (TMS) as internal standard.

Fourier transform infrared spectroscopy (FTIR). FTIR spectra were recorded on a Bruker VECTOR-33 IR spectrometer and the samples were prepared by the KBr disk method.

Gel permeation chromatography (GPC). GPC measurements were conducted at 35 °C on a Waters 1515 by using a series of monodisperse polystyrenes as standard and THF as the eluent with a flow rate of 1.0 mL min⁻¹.

Differential scanning calorimetry (DSC). DSC measurement was performed on a TA Instruments Q2000 differential scanning calorimeter under a nitrogen flow of 50 mL min⁻¹. After the sample was quickly heated to 100 °C and equilibrated at the temperature for 5 min to remove thermal history, it was cooled to -50 °C at a rate of 10 °C min⁻¹. The crystallization temperature (*T_c*) was obtained from the exothermic peak during the cooling scan, and the crystallization enthalpy change (ΔH_c) was calculated from the area under the peak. Each sample was re-heated after cooling down from 100 °C to -50 °C at the same rate to evaluate its melting temperature (*T_m*) and melting enthalpy change (ΔH_m). *T_m* was taken as that corresponding to the maximum heat flow in the re-heating process, and ΔH_m was calculated from the area under the peak. Like in the cases of *T_c* and ΔH_c , we obtained the cold crystallization temperature (*T_{cc}*) and the crystallization enthalpy change (ΔH_{cc}) in the re-heating process.

Polarizing optical microscopy (POM). The morphology of the PU surface was observed by using a Carl Zeiss Axioskop 40 Pol microscope equipped with a digital camera. The film was prepared by spin-casting of a PU solution in THF (0.5 wt%) on a spin coater (CHEMAT, KW-400) at 4000 rpm in air. The film was quickly heated to 100 °C and equilibrated at the temperature for 5 min, then cooled to 25 °C in an oven for 24 h before use.



Scheme 1 Synthesis of CL/GAx macrodiol and CL/GAx-PU.

Table 1 Characterization data of PUs

Sample	Soft segment		M_n^a (g mol ⁻¹)	M_n^b (g mol ⁻¹)	M_w/M_n^b
	Macrodiol	GA content ^a (mol%)			
PCL-PU	PCL	0	2000	1.5×10^4	1.6
CL/GA2-PU	CL/GA2	2	1900	2.0×10^4	1.8
CL/GA6-PU	CL/GA6	6	2000	1.8×10^4	1.7
CL/GA10-PU	CL/GA10	10	2100	1.6×10^4	1.7

^a Determined by ¹H NMR. ^b Determined by GPC.

Contact angle (CA) measurements. The measurements were conducted on a Contact Angle System OCA40 (Dataphysics) at 25 °C. The advancing and receding angles were measured with the sessile drop method by depositing a water drop of 1 μL on the PU surface, then increasing the volume to 4 μL, finally decreasing it. The advancing contact angle and receding contact angle were taken as the maximum and minimum stable angles, respectively. Each CA value was the average of those from five repeated measurements.

Enzymatic degradation. Enzyme-accelerated degradation experiments were performed on an E1 quartz crystal microbalance with dissipation (QCM-D) from Q-Sense AB. The sensor was an AT-cut quartz crystal with a fundamental resonant frequency of 5 MHz, which was mounted in a fluid cell with one side exposed to the solution.²⁸ The effects of surface roughness were minimized by using highly polished crystals with a root-mean-square roughness less than 3 nm.²⁹ The details about QCM-D can be found elsewhere.^{30,31} Briefly, QCM-D simultaneously monitors changes in resonance frequency (Δf) and dissipation (ΔD) in real time. Δf is related to the changes in the mass attached to the oscillating sensor surface, whereas ΔD is to the viscoelasticity of the adsorbed layer. Artificial seawater was used as the reference for degradation measurement.

The Lipase PS solution (0.5 mg mL⁻¹) was delivered to the surface at a flow rate of 150 μL min⁻¹. The changes in frequency (Δf) and dissipation (ΔD) give the information about the degradation and structural change of the PU films. The increase in the oscillation frequency corresponds to the mass loss of the film on the QCM sensor surface. All the presented data were from the third overtone ($n = 3$). Δf and ΔD values from the fundamental were discarded because they were usually noise due to insufficient energy trapping.³² All the experiments were performed at 25 °C.

Hydrolytic degradation. The PU films were prepared *via* a solution casting method. Typically, a solution of 30% (w/v) polymers in THF was dripped onto a glass fiber reinforced epoxy resin panel (20 × 20 mm in size) and kept at room temperature for three days to remove the residual THF. Then, the weight (W_0) of each dried coating together with its panel was measured before dipping into a tank of artificial seawater that was renewed every two weeks. After a certain period of time, the panel was taken out, rinsed three times with deionized water, dried at 60 °C in a vacuum oven, and then the weight (W_1) of the panel was measured again. The difference between W_0 and W_1 is the weight loss of the film, which occurs on the PU surface. For

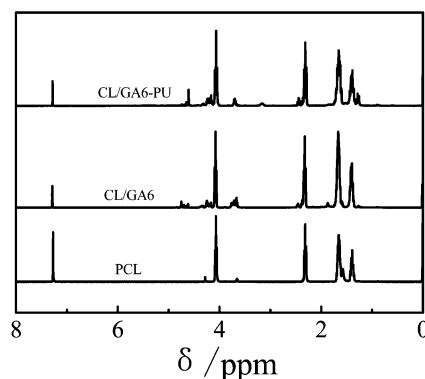
each PU, three coated panels were prepared and measured, and each data point was averaged over three successive and consistent measurements.

Marine field tests. The field tests were performed from May to November, 2012 at the inner Xiamen bay (24°45'N, 118°07'E) in China, where the marine biofouling is heavy because of the rich fouling organisms. The samples applied onto the glass fiber reinforced epoxy resin panels (300 × 250 × 3 mm³) were lowered into seawater at depths of 0.2–2.0 m. After a certain period of time, the panels were taken out of the sea and carefully washed with seawater and photographed, and then they were immediately placed into the seawater to continue the test. Panels coated with CL/GAx-PU were tested, where the control panel consisted of glass fiber reinforced epoxy resin. A set of panels coated with PU carrying DCOIT (10 wt%), a relatively environment-friendly antifoulant, were also examined.

Results and discussion

Fig. 1 shows typical ¹H NMR spectra for PCL, CL/GA6 and CL/GA6-PU. In comparison with PCL, CL/GA6 has a new peak ($\delta = 4.7$ ppm) which is attributed to COCH₂O in GA monomer, indicating the copolymerization of GA and CL. The peaks at 1.25 ppm assigned to the methyl groups in LDI are observed in the spectrum of CL/GA6-PU, indicating the successful synthesis of the polyurethane.

Fig. 2 shows the typical FTIR spectra of CL/GA6 and CL/GA6-PU. The broad band at 3500 cm⁻¹ for CL/GA6 is attributed to the stretching vibration of hydroxyls (OH). For CL/GA6-PU, the

**Fig. 1** ¹H NMR spectra of PCL, CL/GA6 and CL/GA6-PU.

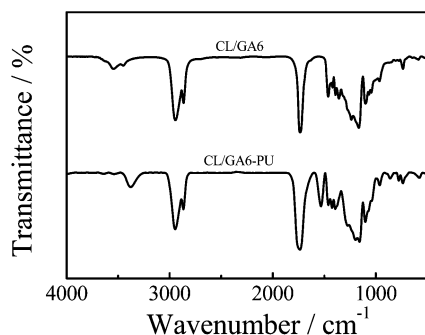


Fig. 2 FTIR spectra of CL/GA6 and CL/GA6-PU.

band at 3300 cm^{-1} corresponds to the stretching vibration of urethane amide bond, and the band for OH is absent. These facts further indicate the synthesis of the polyurethane.

Fig. 3a shows the DSC thermograms of PCL-PU, CL/GA2-PU, CL/GA6-PU, and CL/GA10-PU during the cooling scan. We can observe a single exothermic peak at $-14\text{ }^{\circ}\text{C}$ for PCL-PU due to the crystallization of PCL segments. For CL/GA2-PU, CL/GA6-PU or CL/GA10-PU, the crystallization shifts to a lower temperature. Meanwhile, the exothermic peak area (ΔH_c) becomes smaller as GA content increases (Table 2). When the GA content in the soft segment increases to 10 mol% (CL/GA10-PU), the exothermic peak even disappears. Fig. 3b shows the DSC thermograms from the re-heating process. As GA content increases, either T_m or ΔH_m decreases. The endothermal peak no longer appears when GA content is up to 10 mol%. For CL/GAx-PU with GA content less than 10 mol%, we can observe a cold crystallization peak besides the melting peak. This is because the crystallization of PCL segments are not perfect in the cooling scan,^{33,34} and the crystallization further develops

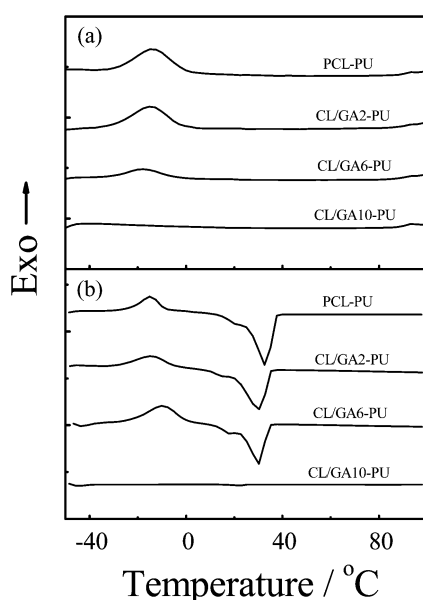


Fig. 3 DSC curves of PCL-PU, CL/GA2-PU, CL/GA6-PU and CL/GA10-PU during (a) the cooling scan, and (b) the re-heating scan.

upon re-heating. Moreover, T_{cc} and ΔH_{cc} increase with GA content, indicating that the crystal irregularity increases. All the facts indicate that the copolymerization of CL with GA reduces the crystallinity of the soft segments. Note that both PCL and PGA can intrinsically crystallize. The decrease of crystallinity is because the copolymerization destroys the regularity of the polymer chains and hinders the crystallization of soft segments.^{35,36}

Fig. 4 shows the POM micrographs of PCL-PU, CL/GA2-PU, CL/GA6-PU and CL/GA10-PU. For PCL-PU, large crystal spherulites and amorphous regions are homogeneously distributed. As GA content in the soft segments increases, the crystalline size decreases. For CL/GA10-PU, it is hard to observe a crystal spherulite. Note that all the PUs have the same hard segments with similar molecular weight and molecular weight distribution (Table 1). Clearly, POM measurements indicate the spherulite size of PU is reduced by the copolymerization of GA and CL. As we know, the degradation rate generally decreases with crystallinity of polyesters in that the crystalline domains are less permeable to water penetration, slowing down the hydrolysis.²¹ On the other hand, the degradation rate increases as the spherulite size decreases because the bigger local crystalline domains are more difficult for water and enzyme molecules to penetrate.^{21,37} Thus, the degradation rate depending on both crystallinity and spherulite size can be moderated by the monomeric ratio.

Fig. 5 shows that either the stationary, advancing or receding contact angle of water on CL/GAx-PU surface is smaller than that on PCL-PU surface. Moreover, as GA content increases, the contact angle gradually decreases, indicating the hydrophilicity increases. The hysteresis between advancing and receding contact angle holds about $32 \pm 4^{\circ}$ for all PUs, indicating that the surface roughness slightly changes.^{38,39} Thus, the effect of surface roughness on the antifouling can be neglected here.

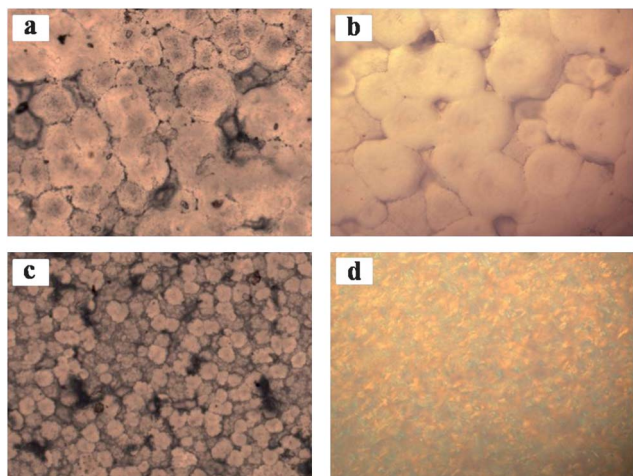
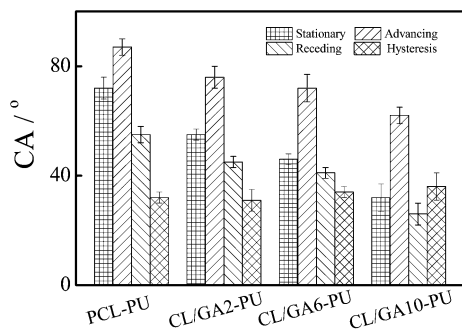
It is known that a large population of microorganisms such as bacteria, actinomycetes and fungi in natural marine environments can produce enzymes.^{21,25} The polyurethane can degrade due to the enzymatic attack by microorganisms present in the seawater. We first examined the enzymatic degradation of the polyurethane by use of QCM-D. Fig. 6 shows the time dependence of frequency shift (Δf) and energy dissipation shift (ΔD) for the enzymatic degradation of the PUs with different GA contents in artificial seawater. For all the PUs, after lipase PS is introduced, Δf sharply increases and then gradually levels off. Upon rinsing with seawater, Δf exhibits a marked increase in comparison with the baseline for seawater. It is known that the increase in mass on the sensor surface causes the frequency to decrease.^{28,31} The sharp increase in Δf indicates the mass loss of the PU films. Namely, some PU chains degrade into small molecules which disperse into the solution. Clearly, the degradation rate increases with GA content, that is, CL/GA10-PU exhibits the largest degradation rate.

In principle, ΔD increases with the thickness but decreases with the rigidity of a layer.^{31,37} Namely, a dense and rigid structure leads to a small dissipation of energy, while a looser structure results in a larger dissipation. For CL/GA6-PU or CL/GA10-PU, the decrease in ΔD indicates the decrease of the film

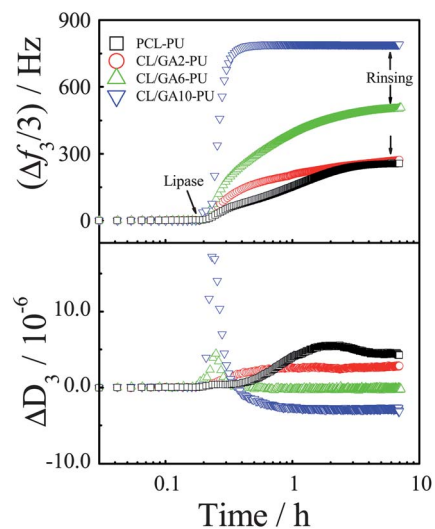
Table 2 Summarized DSC results for PUs

Sample	T_c^a (°C)	T_{cc}^b (°C)	T_m^c (°C)	ΔH_c^a (J g ⁻¹)	ΔH_{cc}^c (J g ⁻¹)	ΔH_m^c (J g ⁻¹)
PCL-PU	-14.0	-14.7	34.3	-23.3	-15.6	44.1
CL/GA2-PU	-15.3	-13.8	31.5	-16.8	-21.4	42.2
CL/GA6-PU	-17.9	-9.6	30.6	-8.2	-29.2	39.2
CL/GA10-PU	—	—	—	—	—	—

^a Determined from the cooling process. ^b Cold crystallization temperature. ^c Determined from the re-heating process.

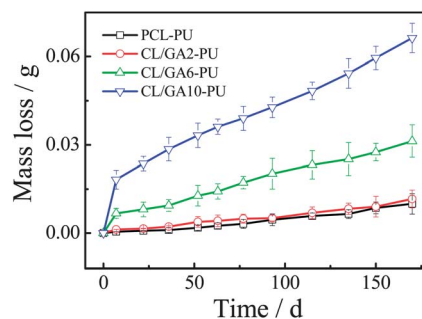
**Fig. 4** The polarizing optical microscopy images of (a) PCL-PU, (b) CL/GA2-PU, (c) CL/GA6-PU, and (d) CL/GA10-PU.**Fig. 5** The contact angle (CA) of water on PCL-PU, CL/GA2-PU, CL/GA6-PU or CL/GA10-PU surface.

thickness due to degradation. In contrast, when the GA content is low enough, *e.g.*, for PCL-PU and CL/GA2-PU, ΔD increases with time. As discussed above, the PCL soft segment can readily crystallize to form larger crystal spherulites with amorphous regions. The enzyme degradation first happens at amorphous regions and then gradually extends to crystal spherulites, where the former is faster than the latter.²¹ The non-uniform degradation makes the film surface much more rougher, leading ΔD to increase. It is understandable that CL/GA2-PU exhibits a behavior similar to PCL-PU because of its low GA content. For CL/GA6-PU and CL/GA10-PU, the copolymerization of CL and GA makes the crystallinity and the spherulite size of soft segments low enough, and the crystallites are small and

**Fig. 6** Time dependence of frequency shift (Δf) and dissipation shift (ΔD) for the enzymatic degradation of the PUs with different GA contents in artificial seawater at 25 °C.

randomly distributed in the amorphous matrix. As a result, the degradation at different sites occurs almost simultaneously, and the thickness and morphology of the film vary evenly.

We also studied the hydrolytic degradation of the PUs in artificial seawater with pH and salt concentration the same as those of natural seawater so that the effects of uncertain factors such as microorganism adsorption are eliminated. Fig. 7 shows the time dependence of mass loss of PU with different GA contents in artificial seawater at 25 °C. After immersing in artificial seawater for 7 days, all the films tested start to lose weight. In comparison with PCL-PU, the PU with CL/GAx soft

**Fig. 7** Time dependence of mass loss of PUs with different GA contents in artificial seawater at 25 °C.

segments generally loses their weight faster, indicating a faster degradation. In addition, the degradation rate of PU increases with GA content. As we know, the hydrolysis rate of polyesters is affected by their molecular structure, ester group density, crystallinity and spherulite size. PCL has much lower degradation rate due to its relatively hydrophobic character and high crystallinity. The copolymerization of CL and GA reduces the crystallinity and spherulite size, so the hydrolysis increases. Moreover, as confirmed by CA measurements (Fig. 5), CL/GAx-PU with a higher GA content has a higher hydrophilicity in that GA units with more ester groups are more hydrophilic than CL units, so its degradability in artificial seawater increases with GA content.

The anti-biofouling of degradable PU was examined by marine field tests. For the control panels whose surfaces are constructed by epoxy resin (Fig. 8a), juvenile barnacles already grow on the surface in 30 days, and the test panels were fully encrusted by 90 days of exposure in marine environments. For the panels coated with degradable PCL-PU (Fig. 8b), the surface is clean within 30 days. Some marine organisms are already located on the surface in 90 days, and it was heavily fouled in 150 days. Fig. 8c shows the panels coated with CL/GA10-PU with a higher degradation rate. The surface remains clean within 30 days and has reasonable anti-biofouling within 90 days. After 150 days, the surface is fouled, but the fouling is less than that regarding PCL-PU. Regardless, the anti-biofouling ability of the degradable PU increases with the degradation rate in the range of GA content 0–10 mol%.

We reported that acrylate copolymers with poly(ethylene oxide-*co*-ethylene carbonate) (PEOC) side chains exhibit good

anti-biofouling in marine environments.²⁰ However, because the side chain is both protein resistant and degradable, the role of degradation in anti-biofouling is not exactly clarified. The present study clearly demonstrates that only degradation can also lead to the anti-biofouling in that the polyurethane does not have protein resistant moieties. When the degradable PUs are immersed into seawater, the seawater and enzymes produced by marine microorganisms makes the ester linkage of polyester cleave and form a self-renewed surface, which can polish the attached living organisms or inorganics. In other words, the degradable surface is dynamic and the attached living organisms or inorganics can be detached by the degradation. Because a higher degradation rate leads to a more dynamic surface with quicker self-polishing, the anti-biofouling ability improves with the degradation rate. However, the PU with GA content in soft segments higher than 10 mol% has a poor adhesion and it would detach from the panel in marine environments. Particularly, a faster degradation results in a shorter service life of the anti-biofouling material while it improves its anti-biofouling ability.

To construct a system with better antifouling ability and longer duration, we added DCOIT, a relatively environment-friendly antifoulant to the degradable PU. Fig. 8d shows the panels coated with PCL-PU containing DCOIT. The surface remains clean within 90 days and still has anti-biofouling after 150 days. As expected, it exhibits much better anti-biofouling ability than PCL-PU without DCOIT. Its antifouling ability is also better than the degradable surface with protein resistance constructed by acrylate copolymers with (PEOC) side chains²⁰ or that by hydrogel.⁴⁰ This is because the PUs can be gradually peeled away in seawater, and the antifoulant buried in the

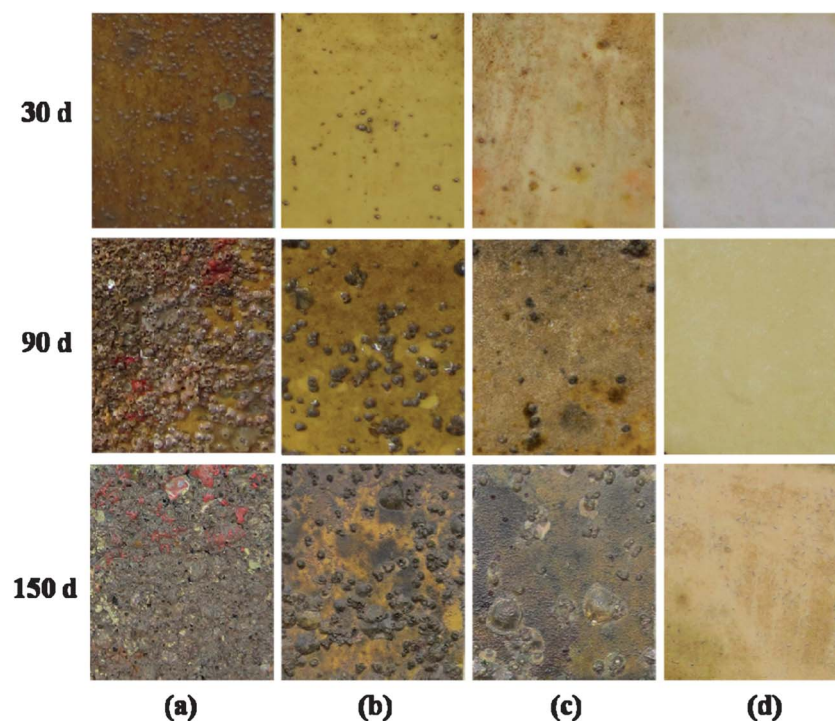


Fig. 8 Typical images of (a) control panel without coating, (b) panels coated with PCL-PU, (c) panels coated with CL/GA10-PU, (d) panels coated with PCL-PU in combination with DCOIT after immersion in seawater (May–October 2012, Xiamen, China) as a function of time.

polymer are gradually released *via* the enzymatic or hydrolytic degradation. Actually, we find that PUs with higher GA content containing DCOIT exhibit the similar anti-biofouling behavior. Namely, the anti-biofouling of the PUs with the antifoulant is independent of the degradation rate in the time we investigated. Thus, we can use the PU with lower degradation rate so that it has both good anti-biofouling and long duration.

It is known that the self-polishing or peeling of the traditional acrylate copolymer with hydrolysable ester group in the side chain largely depends on the seawater motion or sailing speed of a ship.^{1,41} Such polymers generally exhibit a poor anti-biofouling in the case of slow vessels or those in port for long time. Here, the polyurethane with polyester soft segments can degrade due to the hydrolysis of ester linkage and enzymatic attack by micro-organisms in marine environment, which is independent of ship speed. Accordingly, the system exhibits good anti-biofouling ability in both static and dynamic marine environments.

Conclusion

We have prepared degradable polyurethane with ϵ -caprolactone (CL) and glycolide (GA) copolyester oligomer as the soft segments by a combination of ring-opening polymerization and condensation reaction. The copolymerization of CL and GA can reduce the crystallinity and spherulite size of the polyurethane as a function of their monomeric ratio. Enzymatic and hydrolytic degradation experiments demonstrate that the introduction of GA can accelerate the degradation of the PU. Marine field tests show that the degradable polyurethane film with self-renewing surface has good antifouling property. The incorporation of antifoulant can improve the anti-biofouling ability and the service life of the polyurethane. The degradable polyurethane has promise to be developed into long term anti-biofouling material.

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